

Aarush Narang

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EDUCATION

University of Arizona, *Master's in Data Science*

Aug. 2025 - May 2027

Shiv Nadar University, *Bachelor of Technology in Computer Science*

Aug. 2020 - May 2024

EXPERIENCE

Software Engineer, *HCLTech*

July 2024 - July 2025

- Reduced daily support tickets by approx. 26% (2,300 to 1,700) and resolution time by approx. 76% (105 min to 25 min), by building custom copilots in Cursor and Claude Code and shipping via CI/CD; applied responsible AI practices by validating outputs and checking data quality before deployment.
- Cut support tickets approx. 26% and resolution time approx. 76% by integrating an AI-native custom copilot with SharePoint and TypeScript for contextual search across 4,000+ knowledge files; wrote automated tests to validate search behavior and participated in sprint planning and code reviews.
- Enabled scalable employee data management for safety audit and operations workflows by integrating ServiceNow with custom copilots built using ITSM and Claude Code for finance and operations teams; drove process improvement for internal workflows.

Software Engineer Intern, *HCLTech*

Jan. 2024 - July 2024

- Improved visibility of sales targets and deal closure for HR, PMO, and L2 leadership, by developing the Sales KPP web application end-to-end following SDLC: React and TypeScript frontend, C# and .NET backend, and MS SQL Server.
- Delivered data visualization and dashboarding for sales revenue and annual deal closure, by implementing analytics and tracking using React, TypeScript, C#, .NET, and MS SQL Server; collaborated in Agile sprints with senior engineers.

Web Developer Intern, *Carrier*

May 2023 - July 2023

- Drove process improvement by reducing support queries approx. 38% (450 to 280/week) and boosting user engagement 35%, by integrating an AI-native FAQ chatbot into Account Payable Management System and Ticket Tracker and adding automated tests for critical user flows.
- Designed and developed Account Payable Management System and Ticket Tracker for the sales team using HTML, CSS, JavaScript, jQuery, and jqWidgets within the team's SDLC and version control workflow.

Technical Intern, *PricewaterhouseCoopers*

May 2022 - Sep. 2022

- Improved data retrieval speed and system scalability to manage 1,000+ employee records, by building a Graph Database Model with Python and Neo4j and extracting raw data from CNCF to support operations analytics.
- Gained hands-on experience with Golang, Python, Neo4j, and REST APIs while supporting financial data and workflow automation; used Git for version control and followed Agile-style iterations.

PROJECTS

Reflex — Collision Risk Detection (*Python, PyTorch, YOLO11, OpenCV, TensorRT, FastAPI, React, TypeScript, PostgreSQL*)

- Developed a Python and PyTorch system that scans traffic video and flags dangerous close calls between cars, cyclists, and pedestrians before crashes happen, using YOLO11 and OpenCV to detect and track each road user at 93.7% accuracy across 12 hours of footage and 40,000+ road users.
- Built a risk scoring method that flags near misses at 87.6% precision against hand labeled events, logging each in PostgreSQL, and optimized the models with TensorRT to cut processing time 3.8x (12 to 46 frames per second) for live feeds.
- Designed a React and TypeScript dashboard on a FastAPI backend that replays flagged events and maps where close calls cluster, revealing a city's riskiest intersections to guide where to act first.

RadarMD — Medical Imaging Triage System (*Python, PyTorch, MONAI, timm, Grad-CAM, ONNX, FastAPI, Docker, GCP*)

- Developed a deep learning system detecting and localizing 14 thoracic pathologies on chest X-rays, training DenseNet-121 and ConvNeXt with PyTorch Lightning, MONAI, and timm on NIH ChestX-ray14 (112,000+ images, 30,000 patients); reached 0.84 mean AUROC, a 6 point gain over baseline, across 60+ Weights & Biases experiments.
- Implemented Grad-CAM localization to flag regions of concern, validated against 880 ground truth bounding boxes, and built a scikit-learn and torchmetrics test suite prioritizing serious findings over raw accuracy to keep missed diagnoses under 8%.
- Deployed to production via ONNX export for 4x faster CPU inference (~120 ms/image), serving a FastAPI and Gradio app in Docker on Google Cloud Run with GitHub Actions CI/CD, automated tests, and Prometheus health metrics.

Self-Correction for Human Parsing Model (*Python, Git*)

- Benchmarked the SCHP human parsing model against state-of-the-art methods across Look into Person (LIP), Active Template Regression, and Pascal Part, addressing label noise challenges.
- Achieved 84.7% mIoU on the LIP dataset, 2.3% higher than the result reported in the original SCHP paper.

RESEARCH WORK

- Presented the paper titled "Analyzing the Efficacy of Large Language Models: A Comparative Study" with a team of two co-authors at the *35th International Conference on Database and Expert Systems Applications (DEXA 2024)*, held at Naples, Italy (Aug'24). [Paper Link](#)

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL, Go

AI/ML: PyTorch, TensorFlow, MONAI, YOLO11, OpenCV, ONNX, TensorRT, Grad-CAM, Cursor, Claude Code

Web Development: C#, .NET, React.js, Node.js, TypeScript, FastAPI, HTML, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB

DevOps / Cloud: Docker, GCP, GitHub Actions, CI/CD